

Compounds of Transitional elements

221. A clock spring is heated to redness and then plunged into cold water. This treatment will cause it to become

- (a) Soft and ductile
- (b) More springy than before
- (c) Strongly magnetic
- (d) Hard and brittle

222. Mark the steel in which carbon % is highest

- (a) Mild steel
- (b) Hard steel
- (c) Alloy steel
- (d) None of these

223. Mark the variety of iron which has highest melting point

- (a) Pig iron
- (b) Cast iron
- (c) Wrought iron
- (d) Steel

224. Bessemer converter is used in the manufacture of

- (a) Pig iron
- (b) Steel
- (c) Wrought iron
- (d) Cast iron

225. Steel contains

- (a) $Fe + C + Mn$
- (b) $Fe + C + Al$
- (c) $Fe + Mn$
- (d) $Fe + Mn + Cr$

226. Steel differs from pig iron and wrought iron in that it contains

- (a) No carbon at all
- (b) Less carbon than either
- (c) More carbon than either

(d) An amount of carbon intermediate between two

227. Finely divided iron combines with CO to give

- (a) $Fe(CO)_5$
- (b) $Fe_2(CO)_9$
- (c) $Fe_3(CO)_{12}$
- (d) $Fe(CO)_6$

228. Mohr's salt is

- (a) $FeSO_4 \cdot 7H_2O$
- (b) $Fe(NH_4)SO_4 \cdot 6H_2O$
- (c) $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$
- (d) $[Fe(NH_4)_2](SO_4)_2 \cdot 6H_2O$

229. Mohr's salt is

- (a) Normal salt
- (b) Acid salt
- (c) Basic salt
- (d) Double salt

230. An example of double salt is

- (a) Bleaching powder
- (b) $K_4[Fe(CN)_6]$
- (c) Hypo
- (d) Potash alum

231. The passivity of iron in concentrated nitric acid is due to

- (a) Ferric nitrate coating on the metal
- (b) Ammonium nitrate coating on the metal
- (c) A thin oxide layer coating on the metal
- (d) A hydride coating on the metal

232. The action of steam on heated iron is represented as



- (a) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
 (b) $2Fe + 3H_2O \rightarrow Fe_2O_3 + 3H_2$
 (c) $Fe + H_2O \rightarrow FeO + H_2$
 (d) $2Fe + H_2O + O_2 \rightarrow Fe_2O_3 + H_2$
233. Which metal is used to make alloy steel for armour plates, safes and helmets
 (a) Al (b) Mn
 (c) Cr (d) Pb
234. Rusting on iron needs
 (a) Dry air
 (b) Air and water
 (c) Distilled water and carbon dioxide
 (d) Oxygen and carbon dioxide
235. Iron when treated with concentrated nitric acid
 (a) Readily reacts
 (b) Slowly reacts
 (c) Becomes passive
 (d) Gives ferrous nitrate
236. An alloy which does not contain copper is
 (a) Solder (b) Bronze
 (c) Brass (d) Bell metal
237. Which one of the following statements shows the correct percentage of carbon in steel, pig iron and wrought iron
 (a) Steel containing less than 0.15% carbon; wrought iron 0.15 to 2.0% carbon; and pig iron over 2% carbon
 (b) Pig iron less than 0.15% carbon; wrought iron 0.15 to 2.0% carbon; and steel over 2% carbon
 (c) Wrought iron less than 0.15% carbon; steel 0.15 to 2.0% carbon; and pig iron over 2% carbon
 (d) Wrought iron less than 0.15% carbon; pig iron 0.15 to 2.0% carbon; and steel over 2.0% carbon
238. In the Bessemer and open-hearth process for the manufacture of steel, which one of the following is used for the removal of carbon in part or whole
- | | Bessemer | Open-hearth |
|------------------|----------|--------------|
| (a) Ferric oxide | | Air |
| (b) Air | | Ferric oxide |
| (c) Oxygen | | Scrap iron |
| (d) Air | | Scrap iron |
239. About the basic open hearth process, which statement is wrong
 (a) Limestone is added to the charge
 (b) Phosphorus impurity cannot be removed by this process
 (c) Carbon content of the steel can be uniformly controlled over a series of batches
 (d) Iron scrap can be utilized



240. Which of the following statements is wrong

- (a) Heating to a high temperature and then cooling suddenly, *e.g.* by dipping in water, makes steel hard and brittle
- (b) Steel can be softened by heating it to a high temperature for a prolonged time and then cooling slowly. This is called quenching.
- (c) Tempering of hardened steel is done by heating it to just below red heat at controlled temperature and duration
- (d) Phosphorus impurity makes steel 'cold short'

